5. Find all solutions of the congruences

(*a*) $20x \equiv 4 \pmod{30}$;

$20x \equiv 4 \pmod{30}$ iff $\frac{20x}{4} \equiv \frac{4}{4} \pmod{\frac{30}{(30,4)}}$

$5x \equiv 1 \pmod{15}$

however $5x \equiv 0, 5, 10 \Rightarrow$ no solution.

---

6. How many solutions are there to each of the following congruences:

(*a*) $15x \equiv 25 \pmod{35}$;

$15x \equiv 25 \pmod{35}$ iff $\frac{15x}{5} \equiv \frac{25}{5} \pmod{\frac{35}{(35,5)}}$

$3x \equiv 5 \pmod{7}$

since $(3,7) = 1$ $3x$ runs through

complete set of residue ; $3x \equiv 5 \pmod{7}$

holds iff $x \equiv 4 \pmod{7}$

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7. If a is selected at random from $1, 2, 3, \cdots, 14$, and b is selected at random from $1, 2, 3, \cdots, 15$, what is the probability that $ax \equiv b \pmod{15}$ has at least one solution? Exactly one solution?

- $(a, 15) = 1$ then ax runs complete set of residue $(\bmod 15)$

  therefore $ax \equiv b \ (15)$ has unique solution $(\forall b \in \mathbb{Z})$

  prob $= \frac{8}{14}$ because $\#a$ equals $14 - \lfloor \frac{14}{3} \rfloor - \lfloor \frac{14}{5} \rfloor = 14 - 4 - 2 = 8$

- $(a, 15) = 3$ then $ax \equiv b \ (15)$ is solvable iff $3 \mid b$

  $\Leftarrow$ $ax \equiv b \ (15)$ if $\frac{a}{3} x \equiv \frac{b}{3} \left(\frac{15}{(15,3)} \right)$

  $a' x \equiv b' \ (5)$

  since $\frac{a}{3} = 1, 2, 3, 4$ with $(a/3, 5) = 1$

  $(a/3)x$ runs complete set of residue $(\bmod 5)$

  i.e $ax \equiv b \ (5)$ has solution $\underbrace{x_0 \ (\bmod 5)}_{\text{not unique in mod 15}}$

  $\Rightarrow$ suppose $3 \mid ax$ but $3 \nmid b$ then $ax \not\equiv b \pmod 3$

  $ax \not\equiv b \pmod{15}$

  $$\text{prob} = \underbrace{\frac{\lfloor 14/3 \rfloor}{14}}_{3 \mid a} \cdot \underbrace{\frac{\lfloor 15/3 \rfloor}{15}}_{3 \mid b} = \frac{4}{14} \cdot \frac{5}{15}$$

- likewise if $(a, 15) = 5$

  then $ax \equiv b \ (15)$ solvable iff $5 \mid b$ with multiple solutions $(\bmod 15)$

  $$\text{prob} = \underbrace{\frac{\lfloor 14/5 \rfloor}{14}}_{5 \mid a} \cdot \underbrace{\frac{\lfloor 15/5 \rfloor}{15}}_{5 \mid b} = \frac{2}{14} \cdot \frac{3}{15}$$

$$\Pr(\text{solution}) = \frac{8}{14} + \frac{4}{14} \cdot \frac{5}{15} + \frac{2}{14} \cdot \frac{3}{15} = \frac{73}{105}$$

$$\Pr(\text{unique solution}) = \frac{8}{14} = \frac{4}{7}$$

14. Solve the congruences:

$$x^3 + 2x - 3 \equiv 0 \pmod 9;$$
$$x^3 + 2x - 3 \equiv 0 \pmod 5;$$
$$x^3 + 2x - 3 \equiv 0 \pmod{45}.$$

① $x^3 + 2x - 3 \equiv (x-1)(x^2+x+3)$

$\equiv (x-1)(x^2+x-6)$

$\equiv (x-1)(x-2)(x+3) \equiv 0 \pmod 9$

$\Longleftrightarrow$ $x \equiv 1$

or $x \equiv 2$ or $x \equiv 6 \pmod 9$

② $x^3 + 2x - 3 \equiv (x-1)(x^2+x+3)$

$\equiv (x-1)(x^2+x-2)$

$\equiv (x-1)(x+2)(x-1)$

$\equiv \underbrace{(x-1)^2}\ \underbrace{(x+2)} \equiv 0 \pmod 5$

5로 나눴을 때 나머지가 같음.

$\Longleftrightarrow$ $x \equiv 1$ or $x \equiv 3 \pmod 5$

③ $45 \mid x^3 + 2x - 3 \Longleftrightarrow x^3 + 2x - 3 \equiv 0 \pmod 5$

$x^3 + 2x - 3 \equiv 0 \pmod 9$

$\Longleftrightarrow$ $x \equiv \overbrace{1, 2, 6}^{a_1} \pmod 9$

$x \equiv \underbrace{1, 3}_{a_2} \pmod 5$

Note.

$5 \cdot 2 \equiv 1 \pmod 9$

$9 \cdot 4 \equiv 1 \pmod 5$

mod 9 mod 5

by CRT $x \equiv 5 \cdot 2 \cdot a_1 + 9 \cdot 4\, a_2$

$\equiv 10 a_1 + 36 a_2 \pmod{45}$

$\equiv 1, 28, 11, 38, 6, 33$

2.3 THE CHINESE REMAINDER THEOREM

We now consider the important problem of solving simultaneous congruences. The simplest case of this is to find those x (if there are any) that satisfy the simultaneous congruences

$$\begin{aligned} x &\equiv a_1 \pmod{m_1}, \\ x &\equiv a_2 \pmod{m_2}, \\ &\vdots \\ x &\equiv a_r \pmod{m_r}. \end{aligned} \tag{2.1}$$

***23.** Let the m_j be arbitrary positive integers in (2.1). Show that there is a simultaneous solution of this system if and only if $a_i \equiv a_j \pmod{(m_i, m_j)}$ for all pairs of the indices i, j for which $1 \leqslant i < j \leqslant r$.

$\Rightarrow$ suppose $a_1 \equiv a_2 \pmod{(m_1, m_2)}$ and equation have solution

$m_1 \mid x - a_1$ implies $(m_1, m_2) \mid x - a_1$

$m_2 \mid x - a_2$ $\quad (m_1, m_2) \mid x - a_2$

then $a_1 \equiv a_2 \pmod{(m_1, m_2)}$ (조건)

$\Leftarrow$ let $d = (m_1, m_2)$

then $d \mid a_2 - a_1$ $\Rightarrow$ $m_1 x_1 - m_2 x_2 = a_2 - a_1$

$x = m_1 x_1 + a_1$ $\quad$ since $d \mid a_2 - a_1$

$= m_2 x_2 + a_2$ $\quad$ $\exists\, x_1, x_2 \in \mathbb{Z}$ (Bézout's Identity)

---

let $x \equiv y \pmod{M}$ where $M = [m_1, \ldots, m_{k-1}]$

$x \equiv a_k \pmod{m_k}$

let $d = (M, m_k)$

$= ([m_1, \ldots, m_{k-1}], m_k)$

$= [(m_1, m_k) \ldots (m_{k-1}, m_k)]$

$\Rightarrow$ $M x_0 - m_k x_k = a_k - y$

claim $d \mid a_k - y$

$x = M x_0 + y$

$= m_k x_k + a_k$

$\because$ $(m_1, m_k) \mid a_k - a_1$ $\equiv y \pmod{m_1}$

$\ldots$

$(m_{k-1}, m_k) \mid a_k - a_{k-1}$ $\equiv y \pmod{m_{k-1}}$

32. Find all solutions x of $\phi(x) = 24$.

- $\phi(p^j) = p^j\left(1 - \frac{1}{p}\right)$

- Moreover ϕ is multiplicative.

- $24 = 2^3 \cdot 3^1$

2의 거듭제곱 3의 거듭제곱.

$(3, 1)$

13 ↙ $(1, 0)$

7 ↓ $(2, 0)$

5 ↓ $(1, 1)$

↓ $(0, 0)$

- $\phi(13) = 2^2 \cdot 3^1$

$\phi(7) = 2^1 \cdot 3^1$

$\phi(5) = 2^2$

$\phi(3) = 2^1$

$\phi(9) = 2^1 \cdot 3^1$

$\phi(2) = 1$

$\phi(4) = 2^1$

$\phi(8) = 2^2$

$\phi(16) = 2^3$

$(1, 0)$: 13×4

$13 \times 2 \times 3$

13×3.

$(2, 0)$: 7×8 :

$7 \times 4 \cdot 3$.

$(1, 1)$: 5×9 :

$5 \times 2 \times 9$:

$(0, 0)$: 7×5

$7 \times 5 \times 2$

$(3, 1)$ $8 \cdot 9$

34. Prove that there is no solution of the equation $\phi(x) = 14$ and that 14 is the least positive even integer with this property. Apart from 14, what is the next smallest positive even integer n such that $\phi(x) = n$ has no solution?

otherwise, $\phi(x) = \prod_{p^j | x} p^{j-1}(p-1) = 14$

① $7 | x$ otherwise $7 \nmid \phi(x)$ but $7 | 14$

② $6 | \phi(x)$ because $\phi(7^j) | \phi(x)$ (multiplicative)

$6 | \phi(7^j)$

③ $6 \nmid 14$ therefore $\phi(x) \neq 14$

$\phi(3) = 2$, $\phi(5) = 4$, $\phi(7) = 6$

$\phi(15) = 8$ $\phi(11) = 10$ $\phi(13) = 12$. $\Rightarrow$ 14 is smallest.

$\phi(17) = 16$ $\phi(19) = 18$ $\phi(25) = 20$

$\phi(23) = 22$ $\phi(35) = 24$

---

suppose $\phi(x) = 26$

① $13 | x$ otherwise $13 \nmid \phi(x)$ but $13 | 26$

② $12 | \phi(x)$ because $12 | \phi(13^j)$

$\phi(13^j) | \phi(x)$ (multiplicative)

③ $12 \nmid 26$ but $12 | \phi(x)$ $\phi(x) \neq 26$

36. What are the last two digits, that is, the tens and units digits, of 2^{1000}? of 3^{1000}? (H)

- $\phi(25) = 25 \cdot \frac{4}{5} = 20$
- $a^{20} \equiv 1 \pmod{25}$ for every $(a, 25) = 1$

- $2^{20} \equiv 2^{1000} \equiv 1 \pmod{25}$
- $4 \cdot \underbrace{(-6)}_{\text{unique inverse}} \equiv 1 \pmod{25}$
- by CRT $2^{1000} \equiv 4(-6)$

  $\equiv 76 \pmod{100}$

- $3^{20} \equiv 1 \pmod{25}$
- $3^{2} \equiv 1 \pmod{4}$

  therefore $3^{2000} \equiv 1 \pmod{4}$

  $3^{2000} \equiv 1 \pmod{25}$

- by CRT $3^{2000} \equiv 01 \pmod{100}$

***44.** Prove the following generalization of Euler's theorem:

$$a^m \equiv a^{m-\phi(m)} \pmod{m}$$

for any integer a.

if $m = p_1^{j_1} \cdots p_n^{j_n}$

claim $p^j \mid a^{m-\phi(m)}(a^{\phi(m)}-1)$

Note.

ϕ is multiplicative

$\phi(p^j) \mid \phi(m)$

case ① $p \nmid a$ then $a^{\phi(p^j)} \equiv 1 \quad (p^j)$

$a^{\phi(m)} \equiv 1 \quad (p^j)$

case ② $p \mid a$ then $jk \leq m - \phi(m)$

because $m - \phi(m)$ means

$1, \ldots, m$ 중 m과 서로소가

아닌 원소의 갯수.

and $p^1, \ldots, p^k$ is included in that set.

therefore $p \mid a$ and $p^j \mid a^{m-\phi(m)}$

p^j들은 pairwise

coprime 이므로 $m = p_1^{j_1} \cdots p_n^{j_n} \mid a^{m-\phi(m)}(a^{\phi(m)}-1)$

4. Show that the Carmichael number 561 is composite by showing that it is not a spsp(2).

$m = 561, \quad a = 2$

Step ① $\quad m - 1 = 2^j d \qquad j = 4, \quad d = 35$

Step ② $\quad 2^{32} \equiv 256^4$

$\equiv 460^2$

$\equiv 103 \pmod{561}$

$2^{35} \equiv 263 \pmod{561}$

$\not\equiv \pm 1$

Step ③ $\quad 2^{70} \equiv 166 \not\equiv \pm 1 \pmod{561}$

$2^{140} \equiv 67 \not\equiv \pm 1 \pmod{561}$

$2^{280} \equiv 1 \pmod{561}$

$263, \; 166, \; 67, \; 1 \pmod{561}$

$\cancel{+1}$

-1

7. Some earlier authors called a composite number m a pseudoprime to the base a if $a^m \equiv a \pmod{m}$. To distinguish this definition from the one we adopted (at the end of Section 2.1), call such a number m an *old pseudoprime to base a*. Explain why the set of pseudoprimes to base a lies in the set of old pseudoprimes to base a. Demonstrate that the two definitions do not coincide by showing that $m = 161{,}038$ is an old pseudoprime to base 2, but not a pseudoprime to base 2.

new $\qquad\qquad\qquad\qquad$ old

$$a^{m-1} \equiv 1 \pmod{m} \implies a^m \equiv a \pmod{m}$$

$(\Rightarrow)$ $\quad m \mid a^{m-1} - 1 \quad$ therefore $\quad m \mid a(a^{m-1} - 1)$

$(\Leftarrow)$ $\quad \cdot \quad 161038 \nmid 2^{161037} - 1$

even $\qquad$ even $\qquad$ odd

$\cdot \quad 2^{161038} \equiv 2 \pmod{2 \times 73 \times 1103}$

because ① $\quad 2^{161038} \equiv 0 \pmod{2}$

② $\quad 2^{161038} \equiv 2^{46}$

$\equiv 2 \pmod{73}$

with $\quad 2^9 \equiv 2^{72} \equiv 1$

③ $\quad 2^{161038} \equiv 2^{146}$

$\equiv 2 \pmod{1103}$

with $\quad 2^{29} \equiv 2^{1102} \equiv 2$

therefore $\quad \underbrace{2,\ 73,\ 1103}_{\text{coprime}} \quad$ divides $\quad 2^{161038} - 2$

and $\quad 2^{161038} \equiv 2 \pmod{161038}$

13. Let $f(u)$ be a given function. Suppose that a sequence u_i of real numbers is generated iteratively by putting $u_{i+1} = f(u_i)$. Suppose also that $u_1, u_2, \cdots, u_{17}$ are distinct, but that $u_{18} = u_{11}$. What is the least value of s such that $u_{2s} = u_s$?

residue class mod 7

(no smaller then 11)

11, 18, ... $\Rightarrow$ same result

12, 19 . . . $\Rightarrow$ same result.

13, 20 . . . $\Rightarrow$

each residue class implies diff result because $u_1, \ldots, u_{17}$ are distinct

we need s s.t ① $11 \leq s$

③ $\underbrace{2s}_{\text{same value}} \equiv \underbrace{s}_{\text{same value}} \pmod 7 \longleftrightarrow 7 \mid s$

which is $s = 14$

16. *The Pollard $p-1$ method.* Let $d_n = (2^{n!}-1, m)$. Explain why $d_n | d_{n+1}$ for $n = 1, 2, \cdots$. Show that $d_n > 1$ if m has a prime factor p such that $(p-1)|n!$. Apply this approach to find a proper divisor of 403. What is the least n that yields a factor? What is the least n for which $d_n = 403$?

Claim $d_n \mid d_{n+1}$

- $2^{(n+1)!}-1 = (2^{n!})^{(n+1)}-1$

$$= (2^{n!}-1)(1+\cdots+(2^{n!})^n)$$

- $2^{n!}-1 \mid 2^{(n+1)!}-1$ therefore $d_n \mid 2^{(n+1)!}-1$

$d_n \mid m$

Claim $p \mid m$ and $p-1 \mid n!$ then $1 < d_n$

- if p is odd $2^{n!} \equiv 2^{p-1} \equiv 1 \pmod p$

because $p-1$ divides $n!$

- therefore $p \mid d_n$

---

let $x_n \equiv (x_{n-1})^n$

$\equiv 2^{n!} \pmod{403}$

let $d_n = (x_n - 1, 403)$

$x_1 \equiv 2$ $\quad d_1 = (1, 403) = 1$

$x_2 \equiv 4$ $\quad d_2 = (3, 403) = 1$

$x_3 \equiv 64$ $\quad d_3 = (63, 403) = 1$

$x_4 \equiv 64^4$ $\quad d_4 = (\underbrace{325}_{25\times 13}, \underbrace{403}_{13\times 31}) = 13$

$\equiv 326$

search $403 \mid 2^{n!}-1$

$2^{12} \equiv 1 \ (13)$ order $=12$

$2^5 \equiv 1 \ (31)$ order $=5$

$\underbrace{12 \mid n!}_{n\geq 4}$ and $\underbrace{5 \mid n!}_{n \geq 5}$

$\Rightarrow n \geq 5$